Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/06/2022
PI Name:	Urbaniak, Camilla Ph.D.		
Project Title:	Microgravity Analogs as Proxies for Spaceflight to Validate Biofilm Formation and the Exchange of Genes Between Organisms		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	camilla.urbaniak@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	626-840-6992
Organization Name:	ZIN Technologies, Inc		
PI Address 1:	4800 Oak Grove Dr		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:	NOTE: PI had NASA Postdoctoral Program Fellowship at NASA Jet Propulsion Laboratory until 2019; now with Zin Technologies, Inc., Middleburg, OH, but remains physically at NASA JPL address.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	01/04/2021	**End Date:**	01/03/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Venkateswaran, Kasthuri Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC21K0320		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial adaptation to space is an important area to study as it provides information on how microbes on spacecraft and the microbiome of astronauts will change when exposed to the space environment and how that may impact spacecraft integrity and function, and the health of astronauts. However, due to time and cost constraints of conducting studies on the International Space Station (ISS), scientists are not able to perform as many insightful space biology studies as they would like. For this reason, scientists use microgravity simulators to conduct space biology studies, but on Earth. However, there are very limited studies testing how these simulators compare to the "real" thing and the jury is still out on whether microgravity simulators are actually comparable to experiments performed on the ISS. For this reason, we wish to carry out a study that looks at how three simulated microgravity chambers, commonly used on the ground, compare to what would be observed on the ISS. This study would have a ground and a flight component and would look at two bacterial properties that are significant for short and long duration space missions--biofilm formation and the exchange of genetic material between bacteria.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of NASA's goals is to minimize the health risks associated with extended spaceflight, so it is critical that methods for preventing and treating spaceflight-induced illnesses be developed before astronauts embark upon long-duration space missions. In order for this to happen, it is important to learn how bacteria that play a role in human health and disease are affected by spaceflight, such as biofilm formation and horizontal gene transfer. Comparing these responses to popular and frequently used simulated microgravity devices on Earth, and verifying whether the same outcomes can be achieved, will allow more researchers to partake in space related microbiology and human microbiome work, without the extensive limitations imposed by conducting experiments on the ISS. This ultimately diversifies the field and opens up opportunities for scientists from all backgrounds. In addition, understanding microbial adaptation in the new and unique environment of space may help guide future research questions and studies into microbial adaptations on Earth under changing conditions, such as climate change.
Task Progress:	The following has been accomplished for Year 2: -Science Requirements Document (SRD) completed and released. -Hardware provider placed under contract with regular meetings to commence. -Protocol development: • Developed and optimized protocols for Objective 1, which included growth conditions, concentration of antibiotic to use, and polymerase chain reaction (PCR) (including primer design, testing out cycling conditions, and verifying horizontal gene transfer/HGT did occur through PCR). • Developed growth conditions to be used for Objective 2. • Handling, storage, and sample preparation (before and after International Space Station/ISS) have been established for transcriptomics (Objective 2). • Worked on ISS protocols for confocal microscopy (Obj 2) • Worked on ISS protocols for metabolomics (Obj 2). -Bioinformatics pipeline development: • Establishing a pipeline to detect HGT events between donor and recipient genomes (Obj 1). • Establishing a pipeline for biofilm analysis at the genome and transcriptome level (Obj 2)
Bibliography Type:	Description: (Last Updated: 11/01/2024)
Articles in Peer-reviewed Journals	Tesei D, Jewczynko A, Lynch AM, Urbaniak C. "Understanding the complexities and changes of the astronaut microbiome for successful long-duration space missions." Life (Basel). 2022 Mar 28;12(4):495. Review. https://doi.org/10.3390/life12040495 ; PMID: 35454986; PMCID: PMC9031868 , Mar-2022